

9(a)	<i>any nine from:</i> <i>structural genes</i> 1 code for, non-regulatory / structural, proteins / polypeptides ; 2 named example of structural gene ; e.g. <i>lac Z</i> / <i>lac Y</i> / <i>lac A</i> 3 (proteins associated with) rRNA / tRNA ; 4 (proteins such as) enzyme / named (structural) protein ; <i>regulatory genes</i> 5 code for, regulatory / non-structural, proteins / polypeptides ; 6 named example ; e.g. gene coding for repressor protein / <i>lac I</i> / PIF / correct ref. DELLA protein / gene for transcription factors 7 detail ; e.g. switches genes on or off / ref. gene expression / ref. transcription ; <i>repressible enzymes</i> 8 (generally) produced continuously ; 9 synthesis can be prevented by binding of repressor protein to, specific site / promoter / operator ; 10 named example ; e.g. enzyme involved in tryptophan synthesis <i>inducible enzymes</i> 11 synthesis only occurs when, substrate / inducer, is present ; 12 <i>idea that</i> transcription of the gene only occurs when, substrate / inducer, binds to, transcription factor / repressor protein; 13 named example ; e.g. β galactosidase / lactose permease / transacetylase	9
------	---	----------

Question	Answer	Marks
9(b)	<i>any six from:</i> 1 (TF) can form part of protein complex ; 2 (TF) bind to, DNA / promoter / enhancer ; 3 (so) RNA polymerase binds to promoter ; 4 (so) transcription begins / mRNA synthesised / gene expressed / gene switched on ; 5 (or TF binds to DNA) no transcription / no mRNA synthesised / gene not expressed / gene switched off ; 6 can activate genes in correct, order / time / cells / amount ; 7 <i>ref. to</i> correct (pattern of) development ; 8 described example ; e.g. homeobox genes / hox genes / determine sex 9 allow responses to environmental stimuli ; 10 described example ; e.g. correct genes expressed in response to, very high temperatures / light exposure 11 <i>ref. to</i> regulate cell cycle ; e.g. role in cell cycle checkpoints / apoptosis 12 <i>ref. to</i> cell signalling ; e.g. response to hormones	6